

Smart School Feeding Management System

Features Completed — Project Status Update

Student:	Luis Enrique Okal Dulo (Adm: 169685)
Supervisor:	Allan Vikiru
Partner Organisation:	Webmasters Kenya
Programme:	BSc Informatics and Computer Science, Strathmore University
Stack:	Django REST Framework · React Native/Expo · React.js · PostgreSQL · scikit-learn · M-Pesa Daraja A

Authentication & Accounts

- JWT-based login for all roles (parent, student, kitchen staff, bursar, admin).
- Self-registration (parent/student) still available in the mobile app.
- **Admin-driven account creation:** admins can create student accounts (with starting balance) and parent accounts directly from the dashboard, and link them to each other.
- Real parent↔child relationship (many-to-many “guardians”), so a parent can have multiple children and a child can have multiple guardians.
- Role-based permissions enforced end-to-end (admin / bursar / kitchen / parent / student each see only what they are allowed to).

Parent Features (Mobile App)

- View child's real-time meal account balance.
- Top up balance via M-Pesa STK Push (Daraja API, sandbox-tested).
- View payment / top-up history.
- **View meal consumption history** — what their child ate, cost, date, and which kitchen staff served it.
- **Raise support issues / complaints** (balance, meal quality, technical, other) and track status (open → in progress → resolved) with the admin's reply.
- **Apply for credit** — request a higher spending limit so their child can keep eating on credit if funds run low; this raises an approved overdraft ceiling only, never fakes real balance.
- Low-balance alerts.

Kitchen Staff Features (Mobile + Dashboard)

- Search / look up any student by name or ID.
- **Menu-based meal serving:** pick which food items a student is taking (e.g. Rice, Beans, Sukuma) — price is computed automatically as the sum.
- **Search meal combinations:** named bundles (e.g. “Lunch Special”) auto-select all their items and show the combined price instantly, on top of picking individual items.
- Prevents double-serving the same student twice in one day.
- Enforces balance / credit limits before serving (blocks or allows based on approved overdraft).
- “Served by” is automatically recorded via the logged-in kitchen account's name and email — cannot be spoofed.

Admin Dashboard

- Full overview dashboard: meals served today, KES collected, low-balance students, pending anomaly flags, 7-day collection chart, demand forecast preview.
- Student balances table (all students, sortable by balance).
- **Menu Management**: add / edit / remove priced food items; create and manage named meal combinations.
- **Account Management**: create student & parent accounts, link / unlink guardians to students.
- **Credit Requests queue**: review parent overdraft requests, approve (raises spending limit) or reject.
- **Support Issues queue**: view and resolve parent-raised complaints with notes.
- Meal Distribution log: daily list of every meal served, itemised with cost and server identity.
- Demand forecasting (generate & view predictions).
- Anomaly flag review and resolution.
- CSV reports (payments, meal distribution, anomalies, term summary).

Bursar Dashboard

- Student balances, meal distribution log, demand forecast (same visibility as admin).
- **Credit Requests queue** (view-only — financial oversight; only admin can approve/reject).
- **Support Issues queue** (can resolve, same as admin).
- Anomaly flags and CSV reports.
- Sidebar now shows only the pages relevant to their role (no more confusing access-denied pages).

Machine Learning

- **Demand forecasting** — Linear Regression model predicting daily meal counts and cost for the coming week.
- **Anomaly detection** — Isolation Forest model flagging suspicious payment transactions for bursar/admin review.

Payments

- M-Pesa Daraja STK Push integration (sandbox), tested live end-to-end via an ngrok tunnel including the callback path.
- Idempotent payment callback handling (a retried/duplicate Safaricom callback cannot double-credit an account).
- Overdraft/credit system layered on top of real balance (never fakes real money — only a real M-Pesa top-up increases actual balance).

Quality Assurance

- 77 automated backend tests, all passing.
- Full live end-to-end verification pass completed: parent-child linking, issue reporting reaching the dashboard, credit requests reaching admin/bursar, menu-priced serving syncing across mobile and dashboard, and the real M-Pesa payment flow — all confirmed working together correctly.